Student Performance Analytics System

Project Documentation - Python with AI (EWB Courses)

1. Project Title

Project Title	Student Performance Analytics System
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Roll Number	23L51A0595
Training Program	Python with AI - EWB Courses

2. Objective

Teachers usually need to go through many students' marks by hand to find totals, grades, who passed and who needs help. This is slow and easy to get wrong.

This project reads student data from a CSV file and uses **Python, NumPy and Pandas** to automatically:

- calculate total and average marks for every student
- assign grades using a clear grading rule
- decide pass/fail using a clear condition
- find the highest, lowest and average performance
- analyse each subject and each department
- identify the top-performing students

The result is a clear report that can be understood without reading the code.

3. Technologies Used

Tool	Purpose
Python 3	Main programming language (variables, data types, loops, functions, conditions)
NumPy	Numerical calculations: sum, mean, min, max, argmax, argmin, where
Pandas	Loading the CSV, DataFrame operations, sorting, filtering, groupby
Git & GitHub	Version control and project submission

4. Dataset Description

File: **students.csv** - **20 student records**, **9 columns**, no missing values.

Column	Description
Student_ID	Unique ID (S101 - S120)
Name	Student name
Department	CSE, ECE, IT or EEE
Maths, Python, DBMS, Statistics, English	Subject marks out of 100
Attendance	Attendance percentage

The data is a sample dataset created for this project. It has a realistic mix of toppers, average students and students who fail or have low attendance.

5. Implementation

Step 1 - Prepare the dataset. Created students.csv with 20 students, five subjects and attendance.

Step 2 - Read the data. load_data() uses pd.read_csv() to load the file into a DataFrame. The program prints the number of records, columns, first 5 rows and the count of missing values.

Step 3 - Process the data. add_student_metrics() loops through every student (for ... in df.iterrows()). For each student it converts the marks to a NumPy array and calls:

- calculate_total() - uses np.sum()
- calculate_average() - uses np.mean()
- check_result() - uses np.min() to find the lowest subject mark
- assign_grade() - grading rule using if / elif / else

The new columns Total, Average, Result and Grade are added to the DataFrame. np.where() is used to create the Attendance_Status column.

Step 4 - Perform the analysis.

- class_statistics() - pass/fail counts, class average, highest and lowest using np.argmax() / np.argmin()
- subject_wise_analysis() - average, highest, lowest and pass count for each subject
- department_analysis() - Pandas groupby() for department-wise results
- grade_distribution() - number of students in each grade
- get_top_performers() - students sorted by average and ranked
- get_failed_students() - failed students and the subjects they failed in

Step 5 - Final output. main.py prints every result under a clear heading and saves the complete table to student_results.csv.

Rules used

Average marks	Grade
90 and above	A+
80 - 89	A
70 - 79	B
60 - 69	C
50 - 59	D
40 - 49	E
Below 40	F

Pass/Fail rule: a student **passes** only if they score **40 or more in every subject**. A student who fails any subject gets grade **F**.

Attendance rule: attendance below **75%** is marked as Shortage.

6. Key Features

- Loads and organises data with a Pandas DataFrame
- Reusable functions kept in functions.py
- Loops used to process each student
- NumPy used for all numerical calculations
- Total, average, grade and pass/fail for each student
- Highest, lowest and class average
- Subject-wise analysis with best and weakest subject
- Department-wise analysis
- Grade distribution
- Top 5 performers ranking
- Failed students list with the subjects failed
- Attendance shortage detection
- Results exported to student_results.csv

7. Output / Screenshots

Screenshots of the program output (python main.py):

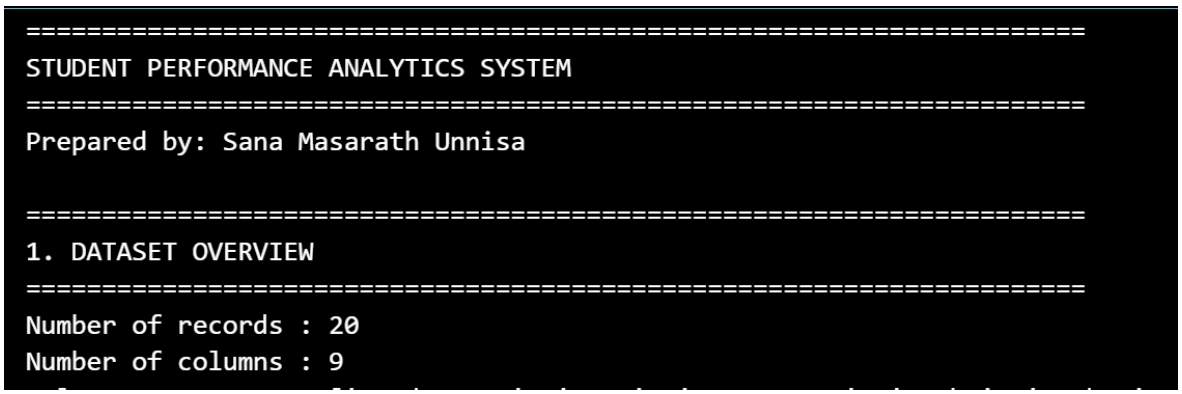


Figure 1: Dataset overview (20 records, 9 columns)

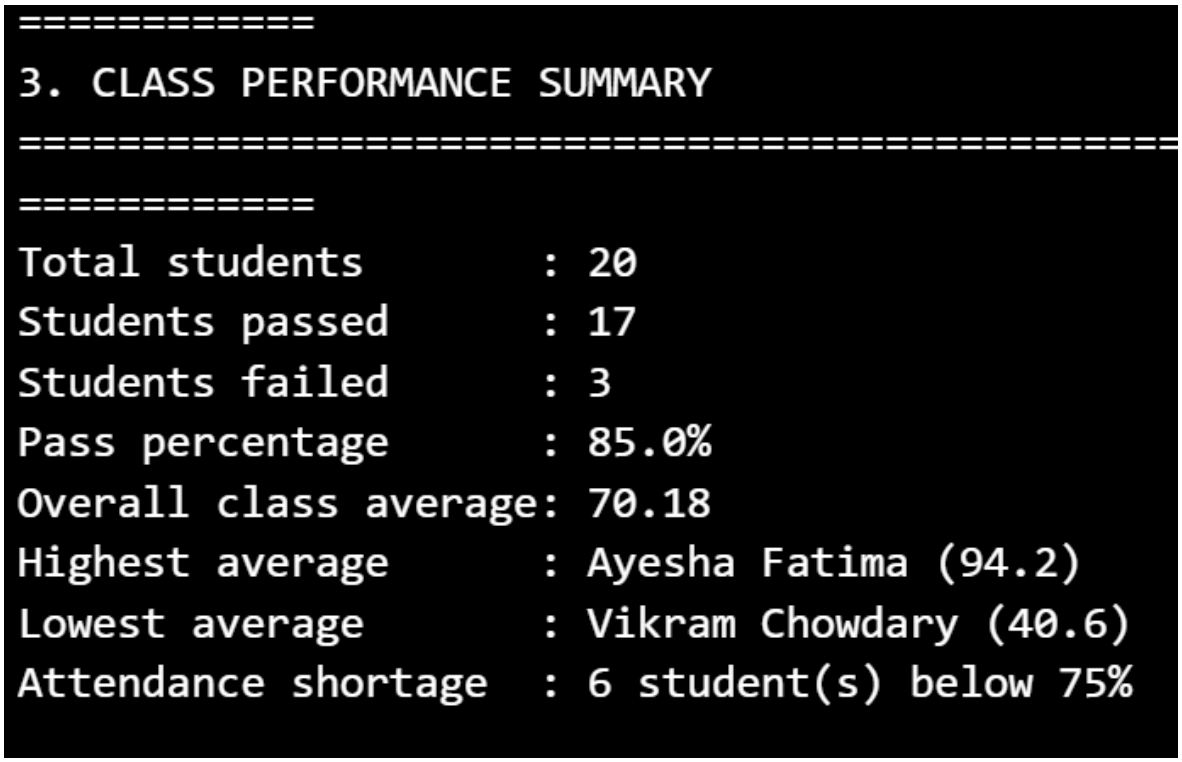


Figure 2: Class performance summary

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4. SUBJECT-WISE PERFORMANCE (best subject first)
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   Subject  Average  Highest  Lowest  Students_Passed
   English    73.15       92      48               20
    Python    71.30       98      38               19
      DBMS    68.90       94      36               19
     Maths    68.80       95      33               18
Statistics    68.75       96      38               19

Best performing subject  : English (average 73.15)
Weakest performing subject: Statistics (average 68.75)
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Figure 3: Subject-wise performance

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5. TOP 5 PERFORMING STUDENTS
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Rank Student_ID             Name Department  Total
Average Grade
   1       S102    Ayesha Fatima        CSE    471
   94.2    A+
   2       S108 Lakshmi Prasanna         IT    453
   90.6    A+
   3       S117     Shreya Patel        CSE    450
   90.0    A+
   4       S101      Aarav Reddy        CSE    433
   86.6     A
   5       S110     Zainab Begum        CSE    430
   86.0     A
```

Figure 4: Top 5 performing students

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6. GRADE DISTRIBUTION
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Grade
A+    3
A     4
B     4
C     4
D     2
E     0
F     3

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Figure 5: Grade distribution

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8. STUDENTS WHO FAILED
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Student_ID            Name Department  Average         Failed_In
    S105     Rahul Verma        EEE     43.6 Maths, Statistics
    S111     Rohit Kumar        ECE     43.4            Python
    S120 Vikram Chowdary        EEE     40.6       Maths, DBMS

=========================================================================
OUTPUT SAVED
=========================================================================
```

Figure 6: Students who failed

8. Final Outcome

The project successfully reads the student dataset and produces a complete performance report. From the 20 records it found:

- **17 students passed and 3 failed** (pass percentage 85%)
- the **class average is 70.18**
- **Ayesha Fatima** has the highest average (94.2) and **Vikram Chowdary** the lowest (40.6)
- **English** is the best-performing subject (average 73.15) and **Statistics** the weakest (68.75)
- **CSE** has the highest department average (85.17) and **EEE** the lowest (49.90)
- 6 students have attendance below 75%

All the required features of the assignment are implemented, and the output can be understood without opening the code.

9. Challenges & Learning

Challenge	How it was solved
A student could have a good average but still fail one subject, which made the grade confusing	Pass/fail is decided by the lowest subject mark (np.min), and any student who fails a subject gets grade F
Deciding where to use a loop and where to use Pandas/NumPy directly	Used a loop for per-student processing and NumPy/Pandas for whole-column calculations
Keeping the code readable as it grew	Moved all calculations into reusable functions in functions.py and kept main.py only for printing the report
Changing rules (pass mark, subjects, attendance limit)	Stored them as constants at the top of functions.py so they can be changed in one place
The program could not find students.csv when run from a different folder	Made main.py build the file path from its own location so it works from any folder

What I learned: working with DataFrames, using NumPy functions such as sum, mean, argmax and where, writing reusable functions, using groupby for summaries, and organising a project so it can be pushed to GitHub and understood by others.

10. GitHub Repository

Link: https://github.com/sanamasarathunnisa-cmd/student-analysis

Project structure

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student-analysis/
  main.py                 - runs the project, prints the analysis
  functions.py            - reusable calculation functions
  students.csv            - student dataset (20 records)
  student_results.csv     - output generated by main.py
  screenshots/            - screenshots of the program output
  Project_Documentation.pdf - project documentation (PDF)
  README.md               - project documentation
```

How to run

```
pip install pandas numpy
python main.py
```